

S M (Rifat) Rafiuddin

Machine Learning & Natural Language Processing Researcher

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Website | Google Scholar | GitHub | ORCID

RESEARCH SUMMARY

Ph.D. Candidate in Computer Science developing efficient, interpretable, and controllable language models through selective computation, mechanistic interpretability, and causal intervention. Builds reproducible PyTorch and Hugging Face experiments for adaptation and evaluation. Record includes **18 peer-reviewed papers and accepted manuscripts, 13 as first or sole author.**

EDUCATION

Oklahoma State University

Stillwater, OK

Ph.D. Candidate in Computer Science; Advisor: Dr. Atriya Sen

Aug. 2022 – Expected July 2027

Qualifying examination passed September 2026.

Dissertation: *Adaptive Computation in Transformers: From Selective Information Processing to Causal Mechanistic Interpretability*

Rajshahi University of Engineering and Technology

Rajshahi, Bangladesh

B.Sc. in Computer Science and Engineering

Jan. 2012 – Oct. 2016

RESEARCH EXPERIENCE

Graduate Research Assistant - rAISON Lab

Summers 2025 and 2026

Oklahoma State University; Advisor: Dr. Atriya Sen

- Developed **MaskLoRA**, using low-rank subspaces to guide token selection; reported experiments achieved **1.3–3.6× speedups** while preserving near-baseline quality (EACL Findings 2026).
- Developed input-conditioned **LoRA rank routing** with **61% fewer adapter FLOPs**; adaptive memory retention preserved **at least 95% of full-model performance** with **30–50% of token representations** in reported experiments.
- Localized transformer controllers for symbolic planning (**91.6%** longest-length accuracy versus **66.3%** for the strongest comparison); developed **C³T** and **CAVE-ABSA** for counterfactual tests and non-target preservation.

Graduate Research Assistant - Complex Systems Lab

Summers 2023 and 2024

Oklahoma State University; Advisor: Dr. Arunkumar Bagavathi

- Developed adaptive contextual masking for aspect-based sentiment analysis (PAKDD 2024).
- Built reproducible pipelines for multimodal fusion, text–image alignment, and model evaluation (ASONAM 2025).

SELECTED FIRST-AUTHOR PUBLICATIONS

- C³T: Counterfactual Causal Reasoning for Sentiment Shifts in Social-Media Conversation Trees.** *EMNLP 2026 Main Conference (accepted; to appear).*
- Context-Conditioned Masked LoRA: Dynamic Rank Routing for Compute-Efficient Parameter-Efficient Fine-Tuning.** *ACL Findings 2026.*
- Emergent Discrete Controller Modules for Symbolic Planning in Transformers.** *ICLR 2026.*
- MaskLoRA: Low-Rank Subspace-Induced Token Masking for Efficient and Faithful Language Models.** *EACL Findings 2026.*
- Learning What to Remember: Adaptive Probabilistic Memory Retention for Memory-Efficient Language Models.** *EMNLP Findings 2025.*
- Constraint-Aware Counterfactual Editing for Aspect-Based Sentiment Analysis.** *HHAI-KEML 2026.*

TEACHING AND ACADEMIC EXPERIENCE

Graduate Teaching Assistant, Oklahoma State University

Stillwater, OK

Computer Science

Aug. 2022 – Present

Supported programming, databases, security, algorithms, and systems courses; lead TA for online Operating Systems (Spring 2026), using Canvas and zyBooks.

Lecturer, University of Asia Pacific and Uttara University

Dhaka, Bangladesh

Computer Science and Engineering

Feb. 2017 – July 2022

Instructor of record in machine learning, algorithms, systems, and programming; mentored published research. UAP evaluations: **25 sections, 647 responses**, weighted **4.12/5** versus a **4.00/5** institutional benchmark.

TECHNICAL SKILLS

Research: Language models, NLP, parameter-efficient fine-tuning, LoRA, mechanistic interpretability, causal and counterfactual reasoning, multimodal learning.

ML Software: Python, PyTorch, Hugging Face Transformers, TensorFlow, Scikit-learn, NumPy, and pandas.

Programming and Tools: C/C++, Java, SQL; Linux, Git, Docker, AWS, and Jupyter.

Systems: GPU experimentation, memory profiling, and training/inference throughput benchmarking.

SELECTED AWARDS AND PROFESSIONAL SERVICE

ICLR 2026 Financial Assistance Award (registration and \$1,600 travel support) | **AIRS Travel Fund (2025)** | **GPSGA Travel and Research Award** (\$900 total, 2024–2025).

Peer review: NeurIPS 2026 Position Paper Track; ICWSM 2026; ACM Transactions on Internet Technology (2026).